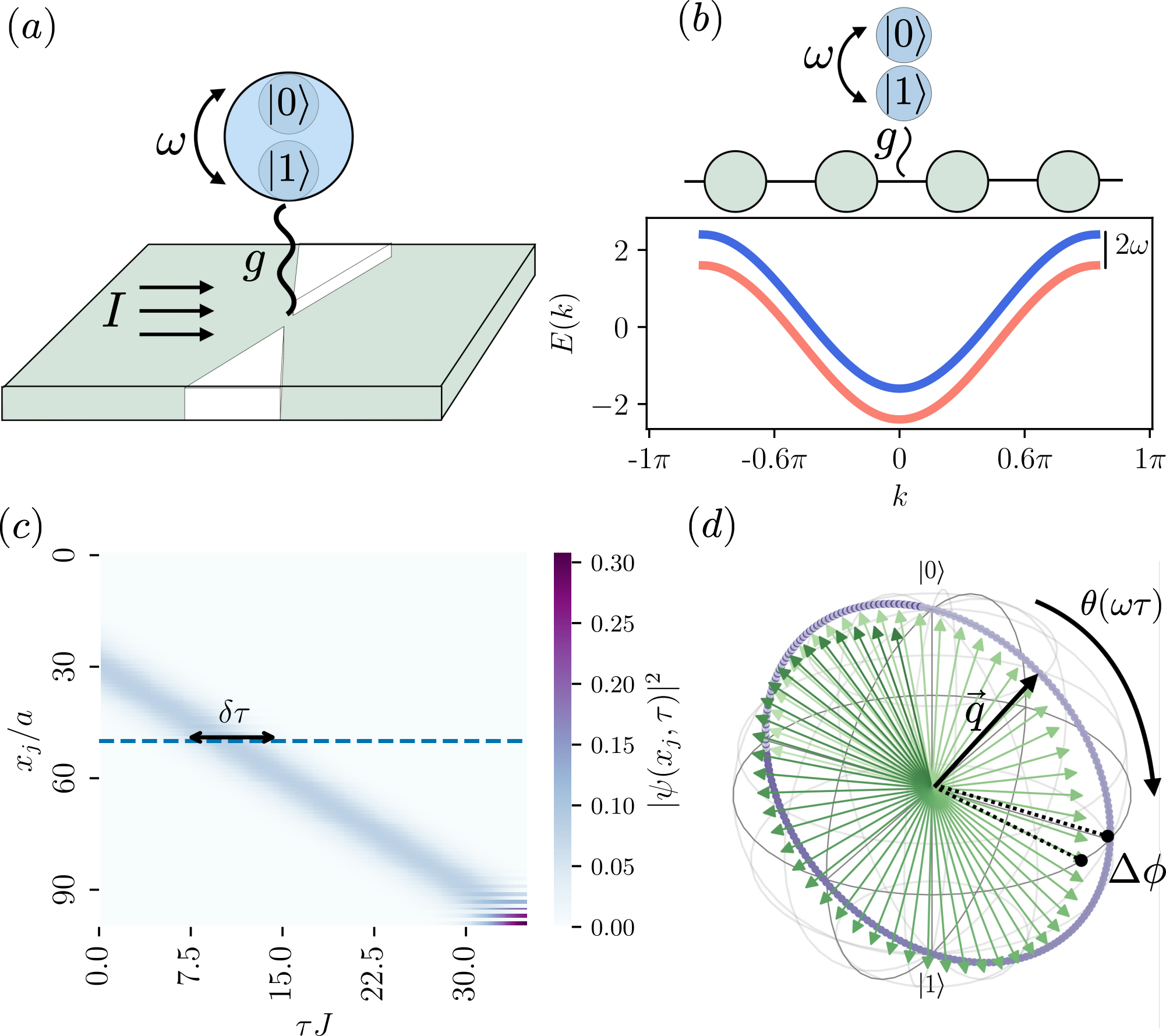

(a)
$\omega$
$|0\rangle$
$|1\rangle$
$g$
$I$
(b)
$\omega$
$|0\rangle$
$|1\rangle$
$g$
$E(k)$
2
0
−2
$2\omega$
-1π
-0.6π
0
0.6π
1π
$k$
(c)
$x_j/a$
0
30
60
90
$\delta\tau$
0.0
7.5
15.0
22.5
30.0
$\tau J$
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$|\psi(x_j,\tau)|^2$
(d)
$|0\rangle$
$|1\rangle$
$\theta(\omega\tau)$
$\vec{q}$
$\Delta\phi$